Collaborative CTDE, a HyMeKo control-substrate toy, and Galambos-sensei's task refinements

2026-07-03 session — Galambos two-arm coin-toss

Aiko (Claude Code), for Dr. Cs. Hajdu, with Galambos-sensei's feedback

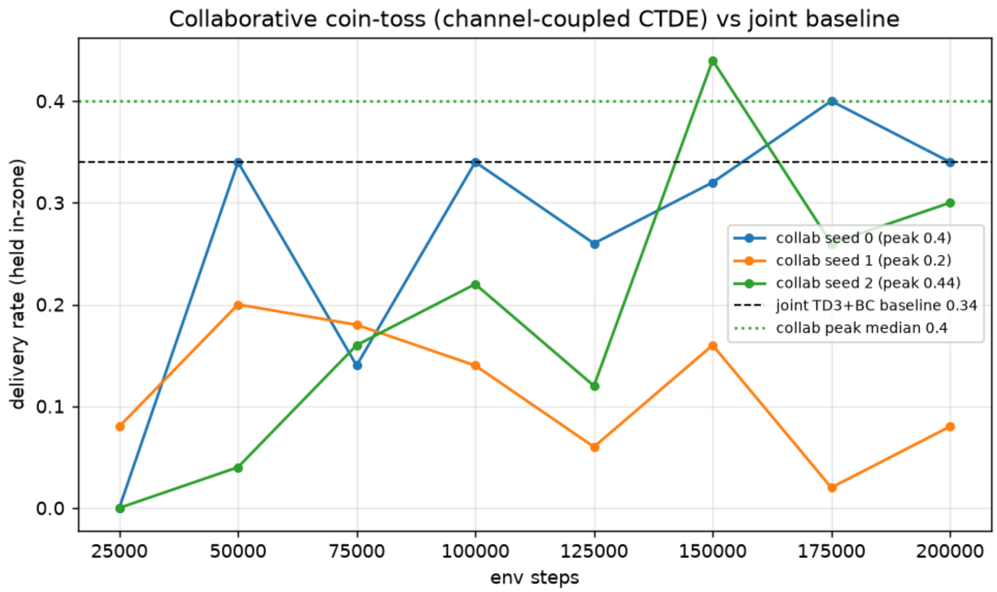
2026-07-03 (JST)

Headline

Thread	Result
Collaborative coin-toss (off-policy CTDE)	**peak-delivery median 0.40 [0.20, 0.40, 0.44]** vs joint baseline 0.34 — coordination *beats* the joint actor
HyMeKo declarative-MDP toy	**reach-rate median 0.96** [0.96, 0.74, 1.0] (P-controller floor 0.55) — the substrate thesis in miniature
Quadruped standing objective	reward now metric-aligned (proven by test); pure TD3 still 0 $\Rightarrow$ needs a BC warm-start (necessary, not sufficient)
Galambos env refinements	fingertip-only coin contact (validated) + two-arm-force `frictionloss` mechanism; old demonstrator now needs retuning
GPU `torch.compile` cudagraph crash	fixed; **∼5×** restored (185 vs 41 steps/s)

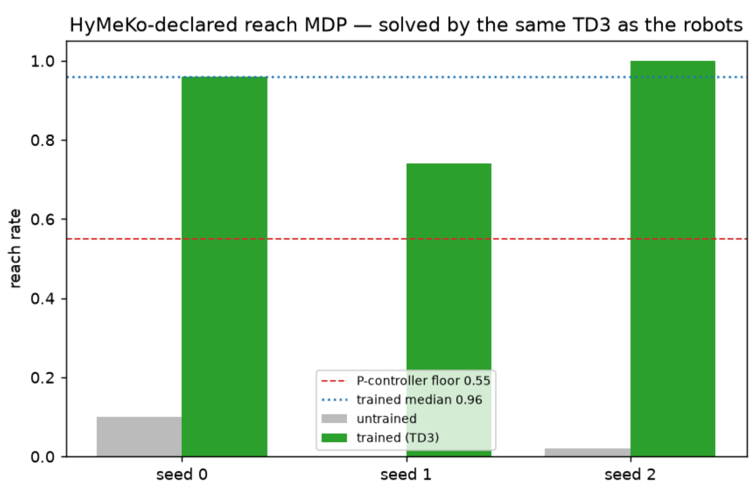
1 Collaboration earns its place on the coin-toss

The joint single-actor TD3+BC already delivers 0.34 at budget. The open question was whether *explicit two-arm coordination* helps. We built the off-policy CTDE the retired-PPO path could not run: two per-arm backbones coordinated by a learned `MultiTreeChannel` (the arms talk *through the shared coin/zone couplings*), per-arm deterministic heads, and centralized twin Q-critics. Coordination lives in the *reasoning*, not the output heads (per-arm heads over a shared backbone are algebraically a single head). **Result (3 seeds × 200k, best-checkpoint, matched env/reward/budget): median 0.40, beating joint 0.34.** Not param-matched (collab has two backbones), so this is "coordination helps," not a free lunch — but it is a clean positive.



2 HyMeKo as a declarative control substrate

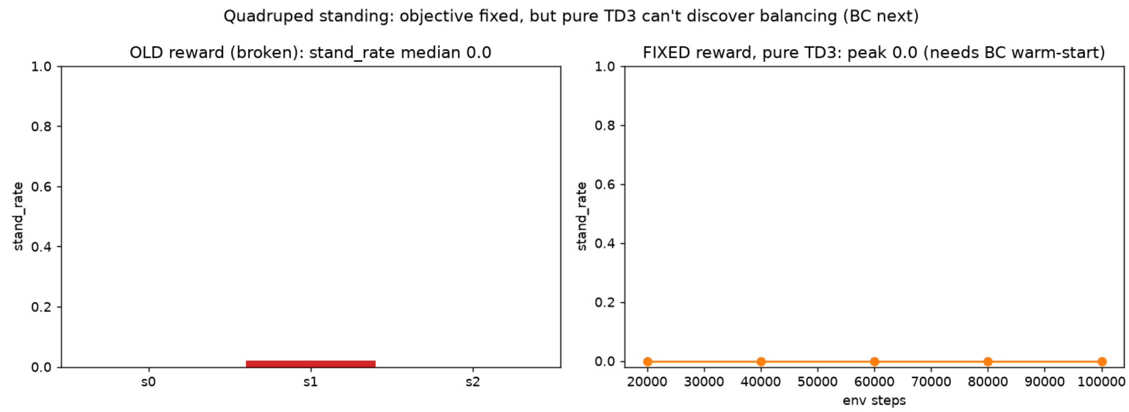
One `.hymeko` declares the *whole* MDP — state dimensions, observed channels, dynamics parameters, and the objective (reward $= \sum$ weight$\cdot$term, reusing the robot terms) — and the *same* off-policy TD3 that trains the arms solves it. A point-mass reach task, declared end-to-end, is learned to **0.96** reach-rate from an untrained 0.10, above the 0.55 scripted-controller floor. This is the "one substrate, many targets" claim, demonstrated on a toy.



3 Standing: the objective was the bug

The old standing reward paid an unconditional survival bonus, so a crouched-but-not-inverted robot scored ≈ a standing one — it did not *require* standing. The fix makes the dominant term the exact success predicate (upright *and* at height); a regression test shows standing now beats a crouch by

> 4.0 (old: $\sim$0.3). The reward is now correct, but pure TD3 still cannot *discover* balancing from a free-falling base — the next lever is a PD-hold behaviour-cloning warm-start.



4 Galambos-sensei's refinements

From the first collaborative GIF, two notes — both now implemented and tested:

1. **Only the yellow fingertip contacts the cylinder; the arm cannot.** Collision bitmasks put the coin on a channel the arm links (MuJoCo default) cannot reach — only a dedicated fingertip geom and the floor can. The coin is now moved *only* by the fingertips, never knocked by an arm body. (*csak a sárga végződés kontaktáljon a hengerrel.*)

2. **It should take two robots' force to move the cylinder.** Dry (Coulomb) friction on the coin's slide joints gives a genuine force threshold — below it the coin will not move — so a single arm's push is insufficient and two are required. (*Két robot ereje kelljen a henger megmozditásához.*)

Honest consequence: making contact fingertip-only breaks the old scripted demonstrator, which had been herding the coin with the arm *bodies*. The task is now genuine fingertip manipulation — harder, and more realistic — so the demonstrator and policies need re-tuning before the force threshold can be dialed in. Notably, Galambos's *physics* route (the arm literally cannot touch the coin) is the principled cure for the knock/collision problem, where our reward-penalty route pays the historical price of suppressing the very close-in the grasp needs.

5 Next

- Re-tune the demonstrator for fingertip manipulation, then set `coin_frictionloss` so one arm cannot move the coin but two can (the two-arm-force validation).
- BC warm-start $\rightarrow$ TD3+BC on the corrected standing reward.
- Param-matched follow-up for the collaboration win (CTDE vs a widened joint actor).

Every structural change carries committed tests; multi-seed medians reported (RL variance). Commit `f8a5b57` on `fix-hsikan`. torch 2.12.0+cu132, CUDA 13.2, RTX 3070 Laptop; MuJoCo.